

Processing Data Stored In Arrays

Generated Aug 18, 2026 7:59 AM

What We'll Learn

By the end of this lesson, students can:

- Process numerical data stored in an array
- Calculate the total and average of array values
- Find the highest and lowest values in an array
- Count array elements that meet a condition
- Combine arrays, loops, and conditions in one program

Before We Start

Consider the following array:

```
int[] scores = {85, 90, 78, 92, 88};
```

We can use a loop to display these values, but what if we need to answer the following questions?

- What is the total of all the scores?
- What is the average score?
- What is the highest score?
- What is the lowest score?
- How many students passed?

Processing array data means examining the stored values and using them to produce useful results.

Let's Understand

An array can store several related values. A loop allows a program to process each value one at a time.

Calculating the Total

Create a variable that begins at 0. Add every array element to it.

```
int[] scores = {85, 90, 78, 92, 88};
int total = 0;

for (int score : scores) {
    total += score;
}

System.out.println("Total: " + total);
```

Output:

```
Total: 433
```

Calculating the Average

The average is calculated by dividing the total by the number of array elements.

```
double average = (double) total / scores.length;

System.out.println("Average: " + average);
```

The value of `total` is converted to `double` so the result can include decimal values.

Finding the Highest Value

The first array element can be used as the starting highest value.

```
int highest = scores[0];

for (int score : scores) {
    if (score > highest) {
        highest = score;
    }
}

System.out.println("Highest: " + highest);
```

Whenever the program finds a value greater than the current highest value, it updates `highest`.

Finding the Lowest Value

The first array element can also be used as the starting lowest value.

```
int lowest = scores[0];

for (int score : scores) {
    if (score < lowest) {
        lowest = score;
    }
}

System.out.println("Lowest: " + lowest);
```

Whenever the program finds a smaller value, it updates `lowest`.

Counting Values That Meet a Condition

A counter can record how many elements meet a particular condition.

```
int passedCount = 0;

for (int score : scores) {
    if (score >= 75) {
        passedCount++;
    }
}

System.out.println("Students who passed: " + passedCount);
```

The counter increases only when the current score is greater than or equal to 75.

Let's Practice

Create a Java program that stores the following daily sales:

```
1200, 950, 1500, 800, 1750
```

Process the array and display:

1. Every daily sales value
2. The total sales
3. The average sales
4. The highest sales value
5. The lowest sales value
6. The number of days with sales greater than or equal to 1000

Expected output:

```
Daily Sales:
```

```
1200
```

```
950
```

```
1500
```

```
800
```

```
1750
```

```
Total Sales: 6200
```

```
Average Sales: 1240.0
```

```
Highest Sales: 1750
```

```
Lowest Sales: 800
```

```
Days with at least 1000 sales: 3
```

Challenge Task

Challenge Title: Student Score Analyzer

Create a Java program that demonstrates the skills learned in Lessons 33–36.

Your program must complete the following requirements:

1. Declare and initialize an integer array containing six student scores.
2. Display all the original scores with their corresponding indexes.
3. Change the score stored at index 2.
4. Display the updated scores using a loop.
5. Calculate and display the total of all scores.
6. Calculate and display the average score.
7. Find and display the highest score.
8. Find and display the lowest score.
9. Count and display how many students received a passing score of 75 or higher.
10. Count and display how many students received a score below 75.
11. Display all scores that are higher than the class average.

Your program output must be clearly labeled and organized.

Example output format:

ORIGINAL SCORES

Index 0: ...
Index 1: ...
Index 2: ...
Index 3: ...
Index 4: ...
Index 5: ...

UPDATED SCORES

Index 0: ...
Index 1: ...
Index 2: ...
Index 3: ...
Index 4: ...
Index 5: ...

SCORE SUMMARY

Total: ...
Average: ...
Highest: ...
Lowest: ...
Passed: ...
Failed: ...

SCORES ABOVE THE AVERAGE

...

How Our Work Will Be Checked

- The array is correctly declared and initialized.
- An array element is correctly accessed and updated.
- Loops are used to traverse the array.
- The total and average are calculated correctly.
- The highest and lowest scores are identified correctly.
- Passing and failing scores are counted correctly.
- Scores above the average are displayed correctly.
- The output is complete, readable, and properly labeled.